

75. A Counterexample for the Local Analogy of a Theorem by Iwasawa and Uchida

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Let $\mathbf{Q}$ be the rational number field, $\bar{\mathbf{Q}}$ the algebraic closure of $\mathbf{Q}$ and k_1, k_2 two finite extensions of $\mathbf{Q}$ contained in $\bar{\mathbf{Q}}$ such that $\text{Gal}(\bar{\mathbf{Q}}/k_1) \cong \text{Gal}(\bar{\mathbf{Q}}/k_2)$ as topological groups. As K. Iwasawa and K. Uchida have independently proved (cf. [2], [6]), k_1 and k_2 are conjugate over $\mathbf{Q}$. In this paper we prove that the analogy for local number fields is not valid: Let p be a prime number, $\mathbf{Q}_p$ the field of p -adic numbers, $\bar{\mathbf{Q}}_p$ the algebraic closure of $\mathbf{Q}_p$. Then there are finite extensions K_1, K_2 of $\mathbf{Q}_p$ contained in $\bar{\mathbf{Q}}_p$ such that $\text{Gal}(\bar{\mathbf{Q}}_p/K_1) \cong \text{Gal}(\bar{\mathbf{Q}}_p/K_2)$ as topological groups, and that K_1 and K_2 are not conjugate over $\mathbf{Q}_p$.

§1. Preliminaries. Let K be a local field, i.e. a commutative field which is complete with respect to a discrete valuation, and L/K a finite Galois extension with $G = \text{Gal}(L/K)$ such that the extension of their residue class fields is separable. Let v_L be the normalized discrete valuation of L , and put $A_L = \{a \in L \mid v_L(a) \geq 0\}$ and $G_x = \{s \in G \mid v_L(s(a) - a) \geq x + 1 \text{ for all } a \in A_L\}$ for $x \geq -1$. The function $\varphi_{L/K}(t)$ for $t \geq -1$ is given by

$$\varphi_{L/K}(t) = \int_0^t \frac{dx}{(G_0 : G_x)}.$$

Let $\psi_{L/K}$ be the inverse function of $\varphi_{L/K}$ and put $G^x = G_{\psi_{L/K}(x)}$. A real number $x \geq -1$ is called a ramification number of L/K (an upper ramification number of L/K , respectively) if $(G_x : \bigcup_{\epsilon > 0} G_{x+\epsilon}) > 1$ (if $(G^x : \bigcup_{\epsilon > 0} G^{x+\epsilon}) > 1$, respectively). When L/K has only one ramification number x , x is also the only one upper ramification number of L/K and vice versa. L/K is totally ramified if and only if $G = G_0$.

Lemma (cf. [4], p. 197 and p. 198). *Let K_i/K be a cyclic extension of degree p with only one upper ramification number t_i for $i=1, 2$, where p is the characteristic of the residue class field of K . Assume that the residue class field extension of K_i/K is separable. Put $M = K_1 K_2$. If $t_1 \neq t_2$, M/K_2 is a cyclic extension of degree p with only one upper ramification number $\psi_{K_2/K}(t_1)$.*

In the above situation, we remark that M/K_2 is totally ramified if K_1/K is totally ramified.

Let ζ_n be a primitive n -th root of 1 in $\bar{\mathbf{Q}}_p$.

Let $K/\mathbf{Q}_p$ be a finite extension contained in $\bar{\mathbf{Q}}_p$ of degree m with residue degree f . Let K_{tr} be the maximal tamely ramified extension of K in $\bar{\mathbf{Q}}_p$. The Galois group of K_{tr}/K is the total completion of a group generated by two elements s, t satisfying a unique relation $s^{-1}ts = t^{p^f}$ (cf. [1], p. 463). Let r be the largest integer such that a primitive p^r -th root of 1 is contained in K_{tr} . Put $q = p^r$. Let c be an integer such that $\zeta_q^c = \zeta_q^s$, and τ the uniquely determined p -adic integer such that $\tau^{p-1} = 1$ and $\zeta_q^c = \zeta_q^t$. We fix c for each K . Then the Galois group $\text{Gal}(\bar{\mathbf{Q}}_p/K)$ is determined by m, f, q, c, τ for odd p and by m, f, q, c (if $q \geq 4$) for $p=2$ as a topological group (cf. [3], [7]). If $K(\zeta_q) = K$, we assume $c=1$ in § 2.

§ 2. Examples. Theorem. (i) Let p be odd. Denote by K_0

$$\mathbf{Q}_p(\sqrt[p]{\zeta_p-1}),$$

and by K_i for $i=1, 2, \dots, p-2$

$$\mathbf{Q}_p(\zeta_p, \alpha_i) \text{ with } \alpha_i \in \bar{\mathbf{Q}}_p \text{ such that } \alpha_i^p - \alpha_i = \left(\frac{1}{\zeta_p-1}\right)^i.$$

Then $\text{Gal}(\bar{\mathbf{Q}}_p/K_i)$ and $\text{Gal}(\bar{\mathbf{Q}}_p/K_j)$ are topologically isomorphic for $0 \leq i, j \leq p-2$, but K_i and K_j are not conjugate over $\mathbf{Q}_p$ if $i \neq j$.

(ii) Denote by L_0

$$\mathbf{Q}_2(\sqrt{\zeta_4-1}),$$

and by L_1

$$\mathbf{Q}_2(\beta) \text{ with } \beta \in \bar{\mathbf{Q}}_2 \text{ such that } \beta^2 - \beta = \frac{1}{\zeta_4-1}.$$

Then $\text{Gal}(\bar{\mathbf{Q}}_2/L_0)$ and $\text{Gal}(\bar{\mathbf{Q}}_2/L_1)$ are topologically isomorphic, but L_0 and L_1 are not conjugate over $\mathbf{Q}_2$.

Proof. (i) By [5], p. 79 and p. 80, $K_i/\mathbf{Q}_p(\zeta_p)$ is a totally ramified cyclic extension of degree p with only one upper ramification number p for $i=0$ and i for $i=1, 2, \dots, p-2$. By [5], p. 86, $\mathbf{Q}_p(\zeta_{p^2})/\mathbf{Q}_p(\zeta_p)$ has the only one upper ramification number $p-1$. Let $m_i, f_i, q_i, c_i, \tau_i$ for K_i be as in § 1. We have $m_i = p(p-1), f_i = 1$. Since by Lemma $K_i(\zeta_{p^2})/K_i$ is totally ramified of degree p , we have $q_i = p$ and so $c_i = \tau_i = 1$. Therefore $\text{Gal}(\bar{\mathbf{Q}}_p/K_i)$ and $\text{Gal}(\bar{\mathbf{Q}}_p/K_j)$ are topologically isomorphic for $0 \leq i, j \leq p-2$ (cf. § 1). On the other hand, for $s \in \text{Gal}(\bar{\mathbf{Q}}_p/\mathbf{Q}_p)$ we have $s(K_0) = \mathbf{Q}_p(\sqrt[p]{s(\zeta_p)-1})$ and $s(K_i) = \mathbf{Q}_p(s(\zeta_p), s(\alpha_i))$ with $s(\alpha_i) \in \bar{\mathbf{Q}}_p$ such that $(s(\alpha_i))^p - s(\alpha_i) = \left(\frac{1}{s(\zeta_p)-1}\right)^i$ for $i=1, 2, \dots, p-2$. Counting the upper ramification number (cf. [5], p. 79 and p. 80), we have K_i and K_j are not conjugate over $\mathbf{Q}_p$ if $i \neq j$. (ii) Similarly $L_i/\mathbf{Q}_2(\zeta_4)$ is a totally ramified quadratic extension with only one upper ramification number 4 if $i=0$ and 1 if $i=1$. By [5], p. 86, $\mathbf{Q}_2(\zeta_8)/\mathbf{Q}_2(\zeta_4)$ has the only one upper ramification number 3. Let m_i, f_i, q_i, c_i for L_i be as in § 1. By the same way as in (i), counting $m_i=4, f_i=1, q_i=4, c_i=1$ we have that

$\text{Gal}(\bar{Q}_2/L_0)$ and $\text{Gal}(\bar{Q}_2/L_1)$ are topologically isomorphic, and that L_0 and L_1 are not conjugate over Q_2 .

Remark. Let K_i be a local field of characteristic $p > 0$ with finite residue class field k_i for $i=1, 2$. Let K_i^{ab} be the maximal abelian extension of K_i for $i=1, 2$. Assume that $\text{Gal}(K_1^{ab}/K_1)$ and $\text{Gal}(K_2^{ab}/K_2)$ are topologically isomorphic. Then K_1 and K_2 are isomorphic.

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